



## **Financial risk modelling using machine learning techniques**

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### 3. Exploratory Data Analysis (EDA)

#### *Quantitative Approach*

According to Hassan (2024), quantitative research designs are characterised by their reliance on numerical data, structured methodologies, and statistical tools to test hypotheses and generate generalisable findings. This approach supports the systematic examination of borrower level financial information, enabling the identification of trends, correlations, and predictive patterns that inform credit risk assessment. The study will use a quantitative design to analyse variables and apply machine learning algorithms to improve the reliability of lending decisions while minimizing the risk of default. By applying machine learning algorithms within this quantitative framework, the study leverages mathematical optimisation and empirical evaluation to develop predictive models capable of improving the accuracy and reliability of credit risk classification.

In this study, the quantitative design is done using a structured datasets, numerical variables, and algorithmic modelling procedures. Machine learning techniques such as logistic regression, random forests, decision tree and gradient boosting are applied to the model. The model will be evaluated using train test splits and the use of performance metrics such as accuracy, precision, recall, F1-score, and the area under the ROC curve (AUC). These metrics provide objective criteria for comparing model performance and determining the most effective predictive technique. The structured, replicable nature of quantitative research ensures that findings can be validated and generalised across similar financial contexts (Lou, 2024).

#### *Populating/Sampling*

This study's target group consists of individuals who use formal financial systems and are subject to credit risk assessment by financial organizations. The dataset includes 5,000 customer records representing borrowers with different kinds of demographic, financial, and behavioral factors. These records consist of individuals with different income levels, credit scores, debt obligations, banking activity, and payback histories, resulting in a diversified sample for credit risk assessment.

This study uses an open-source dataset from Kaggle. It is publicly available and already includes a predefined set of observations produced by the original data supplier. As a result, the author did not perform any extra sample procedures, and the analysis is fully based on the previously released dataset. The study therefore takes a census approach, to analyze 5,000 records in the dataset. Using the whole dataset ensures that all relevant information is kept and avoids the risk of sampling bias caused by selecting only a selection of observations. Prior to model building, the dataset will be separated into training and testing subsets to simplify model training, validation, and performance evaluation.

## Data Collection

The dataset will be downloaded straight from the Kaggle website and imported into the analytical environment for processing and analysis. The study does not collect primary data and instead relies solely on secondary data sources. The data will be collected from a Kaggle open dataset (<https://www.kaggle.com/datasets/programmer3/credit-risk-dataset>). The link attached is now unavailable. The data is now available on GitHub ([https://github.com/Sponono/RPDA8412/blob/baed8f70f9ecbd5e6bc29d9d9a4b78b073998dd6/credit\\_risk\\_dataset.csv](https://github.com/Sponono/RPDA8412/blob/baed8f70f9ecbd5e6bc29d9d9a4b78b073998dd6/credit_risk_dataset.csv))

The dataset will go through a number of preparation steps after data gathering in order to get it ready for machine learning analysis. Data cleansing, data quality checking, addressing inconsistencies, categorical variable encoding, and feature scaling when required are some of these processes. To understand the distribution of variables and find possible connections between borrower characteristics and credit risk outcomes, exploratory data analysis (EDA) will be carried out. Several machine learning algorithms for the grouping and prediction of credit risk will next be trained and assessed using the prepared dataset. In addition to ensuring that a sizable dataset is accessible for model construction and evaluation, the utilization of secondary data provides an economical and effective method for examining borrower credit risk.

## 3.5 Data Analysis

### 3.5.1 Explanatory Data Analysis

The Exploratory Data Analysis (EDA) phase provides an initial understanding of the Credit Risk and Loan Default Analysis Dataset obtained from Kaggle. This phase will help identify data quality issues, understand variable distributions, and uncover patterns that may influence the performance of machine learning models used for credit risk default prediction.

The EDA will begin with data load using Python libraries:

- Numpy (Import as np) for mathematical functions (arrays)
- Pandas (import as pd) for cleaning, filtering, transforming, and analysing data.
- Seaborn (Import as sns) for statistical visualizations
- Matplotlib (Import matplotlib.pyplot as plt) for interactive data visualization
- Split data from sklearn.model\_selection to import train\_test\_split
- Encode data to numerical value (1/0) from sklearn.preprocessing to import OneHotEncoder
- Train model from sklearn.linear\_model to import LogisticRegression
- Train model from sklearn.tree to import DecisionTreeClassifier
- Train model from sklearn.ensemble to import RandomForestClassifier
- Train model from xgboost to import XGBRegressor
- Scaling numerical features from sklearn.preprocessing to import StandardScaler.

- Model training from sklearn.metrics import accuracy\_score, recall\_score, precision\_score, root\_mean\_squared\_error (Müller & Guido, 2022)

### 3.5.2 Data Evaluation

The dataset includes a dependent variable, “Credit\_Risk”, which represents the customer's credit risk classification (low-risk or high-risk). The dataset also has a few independent variables that describe the demographic, financial, and behavioral characteristics of customers. These include “Age”, which is the customer's age; “Gender”, which is the customer's gender; and “Employment\_Status”, which describes the customer's employment situation.

Financial variables include “Annual\_Income”, which is the customer's yearly income; “Credit\_Score”, which measures the customer's creditworthiness based on their credit history; “Outstanding\_Debt”, which shows the total amount of debt the customer owes; “Average\_Balance”, which shows the average balance maintained in the customer's account; “Open\_Loans”, which is the number of active loans held by the customer; and “Loan\_Application\_Amount”, which shows the amount requested in a loan application.

The dataset also has a few behavioral variables that reflect banking and repayment behavior. These include “Mobile\_Banking\_Logins”, which measures the frequency of mobile banking usage; “Online\_Transfer\_Frequency”, which shows how often the customer performs online transfers; “Debit\_Card\_Usage”, which shows the frequency of debit card transactions; “ATM-Withdrawal\_Frequency”, which measures how often the customer withdraws cash from ATMs; and “Late\_Payment\_Count”, which records the number of late payments made by the customer. Additional risk-related variables include “Loan\_Default\_History”, which shows whether the customer has previously defaulted on a loan, and “Fraud\_Flag”, which identifies customers associated with potentially fraudulent activities.

### 3.5.3 Data Cleaning

The dataset will be loaded and evaluated in Python, where an initial assessment is conducted to ensure data quality and readiness for analysis. Missing values are identified to determine the extent of data incompleteness and to inform appropriate handling strategies. The data types of all variables are then verified to ensure that each feature is correctly classified for numerical or categorical analysis. Duplicate records are also detected and removed where necessary to prevent skewed results, biased model training, or inflated performance metrics.

Data Encoding - categorical variables such as ‘gender’ and ‘employment\_status’ are converted into numeric formats using one-hot encoder as shown in Figure 1

Where Female = 0 and Male = 1

And Employed = 1, Self-Employed = 2, Unemployed = 3

|   | age | gender | employment_status | annual_income | account_age_months |
|---|-----|--------|-------------------|---------------|--------------------|
| 0 | 58  | 1      | 3                 | 58137.75119   | 115                |
| 1 | 48  | 1      | 2                 | 26174.92283   | 32                 |
| 2 | 34  | 2      | 3                 | 75566.83726   | 14                 |
| 3 | 62  | 1      | 2                 | 35197.96152   | 179                |
| 4 | 27  | 0      | 2                 | 12136.99835   | 225                |

**Figure 1**

Scaling the Data - numerical features such as age, annual\_income, avg\_monthly\_balance, avg\_deposit\_amount, debit\_card\_spending, total\_outstanding\_debt, loan\_application\_amount are scaled using standardization to ensure that variables with different magnitudes are placed on a comparable scale for algorithms such as Logistic Regression and XG Boost. This process transforms the data to have a mean of 0 and a standard deviation of 1, which helps improve the performance, stability, and efficiency of machine learning models, the code is shown below, Annexure A.

```
#Scaling data for Logistic Regression and XG Boost
preprocess_scaled = ColumnTransformer(
    transformers=[
        ('num', StandardScaler(), num_cols)
    ],
    remainder='passthrough'
)

#Non_Scaling data for Decision Tree and Random Forest
preprocess_noscale = ColumnTransformer(
    transformers=[
        ('num', 'passthrough', num_cols)
    ],
    remainder='passthrough'
)
```

## Annexure A

### Feature Selection

Feature Selection involves identifying and selecting the most relevant features (variables) that contribute to the prediction of the target variable. Correlation analysis and univariate statistical tests such as ANOVA F-tests and Recursive Feature

Elimination (RFE) are be used to identify the most relevant predictors and remove variables that do not meaningfully contribute to the linear regression model.

ANOVA F-Test

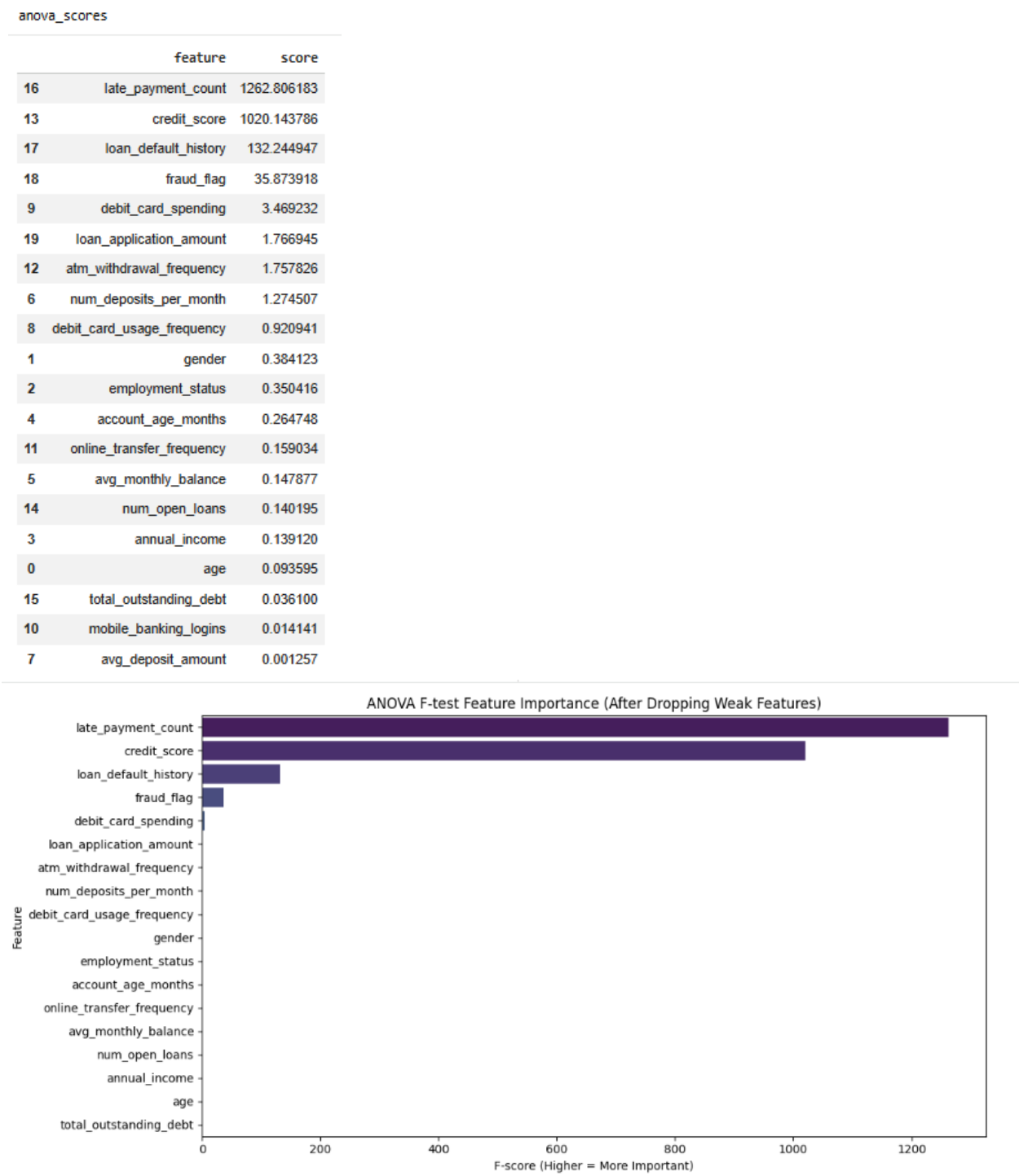


Figure 2

(GeeksforGeeks, 2025)

The ANOVA F-test results above in Figure 2 show that behavioural and financial factors are the strongest predictors of loan default. Late payment count and credit score have very high F-scores, showing clear differences between customers who defaulted and those who did not. Loan default history and fraud flags also show strong statistical significance.

Other factors, such as debit card spending, loan application amount, ATM withdrawal frequency, and the number of monthly deposits, show moderate predictive value. This suggests that customers' financial and transaction behaviour can also help identify loan risk.

In contrast, demographic factors such as gender, employment status, and age have very low F-scores, meaning they have limited predictive power on their own. Overall, the ANOVA results show that behavioural and financial factors provide the strongest differences between customers who default and those who do not (GeeksforGeeks, 2025).

Recursive Feature Elimination (RFE)

```
[ 'gender',
  'employment_status',
  'num_deposits_per_month',
  'online_transfer_frequency',
  'atm_withdrawal_frequency',
  'credit_score',
  'num_open_loans',
  'late_payment_count',
  'loan_default_history',
  'fraud_flag']
```

---

**Figure 2.1**

RFE in Figure 2.1 above identified 10 important factors that can help predict loan default risk. The strongest factors were related to customer behaviour, such as the number of late payments, previous loan defaults, fraud flags, and transaction activity. Financial factors, including credit score and the number of open loans, also had an important influence. Gender and employment status were included because they showed relationships with financial and behavioural patterns. Overall, the selected features provide a realistic and easy-to-understand view of credit risk (Lou, 2024).

### 3.5.4 Logistic Regression

#### Confusion Matrix

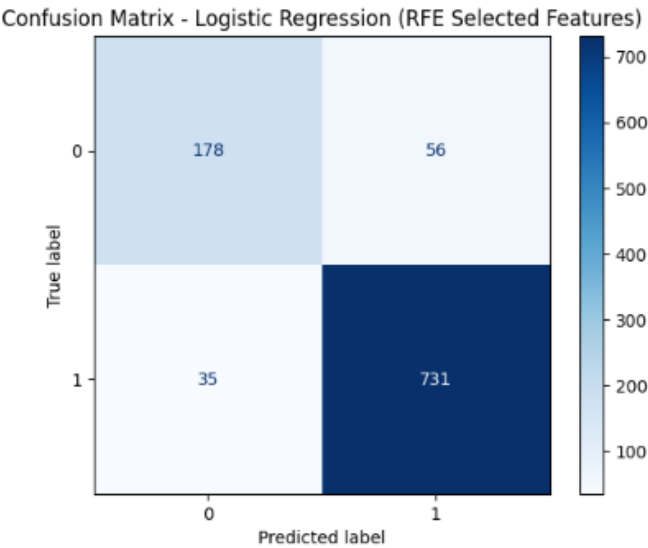


Figure 3.5.4.1

The confusion matrix in Figure 2.2 above shows that the Logistic Regression model performs well in identifying customers who are likely to default. The model correctly identified 731 defaulting customers, while 35 defaulting customers were incorrectly classified as low-risk. These false negatives are important because failing to identify a high-risk customer could result in financial losses.

The model also correctly identified 178 customers who did not default. However, 56 non-defaulting customers were incorrectly classified as high-risk. These false positives could result in missed lending opportunities.

Overall, the results show that the model is effective at identifying high-risk customers, with a reasonable balance between false positives and false negatives.

```
=====
LOGISTIC REGRESSION MODEL EVALUATION
=====

Accuracy Score: 0.909

Classification Report
precision    recall  f1-score   support

   0       0.84     0.76     0.80       234
   1       0.93     0.95     0.94       766

 accuracy          0.91       1000
 macro avg         0.88     0.86     0.87       1000
weighted avg         0.91     0.91     0.91       1000
```

Figure 3.5.4.2

The accuracy score in Figure 3.5.4.2 is 91.2%; this means that the model is reliable and performs very well for a loan default prediction. The precision for low risk is correct 84%

of the time, and high-risk is correct 93% of the time. The model correctly identifies 76% of low-risk customers, and 96% of high-risk customers. The model works well in recognizing low-risk customers and is also highly effective in identifying high-risk customers, however the recall for low-risk cases is lower at 0.90, implying that there are low-risk customers who are mistakenly misclassified as high-risk. From a business perspective, this approach is effective at reducing bad-debt risk because it identifies almost all high-risk accounts. However, it also has a small downside because it incorrectly restricts almost a quarter of good customers (Lou, 2024).

3.5.5 Random Forest  
Confusion Matrix

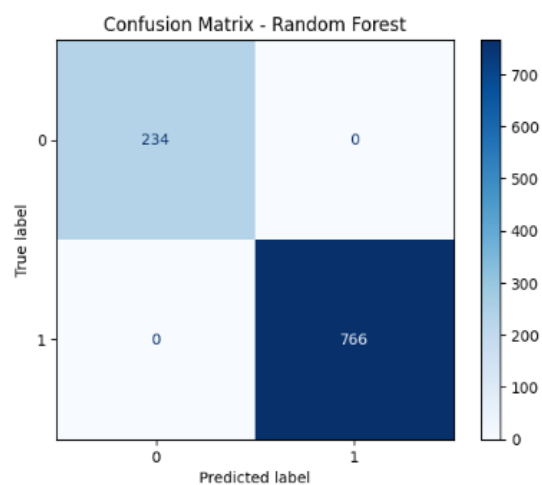


Figure 3.5.5.1

The Random Forest model in Figure 3.5.5.1 achieved perfect accuracy, with no false positives or false negatives. This means the model correctly classified all the test cases. The results suggest that the RFE-selected features provided clear patterns that the group of decision trees could use to make accurate predictions.

```
=====
RANDOM FOREST MODEL EVALUATION
=====

Accuracy Score: 1.000

Classification Report
      precision    recall  f1-score   support

     0       1.00      1.00      1.00        234
     1       1.00      1.00      1.00        766

 accuracy          1.00          1.00          1.00       1000
 macro avg          1.00          1.00          1.00       1000
weighted avg          1.00          1.00          1.00       1000
```

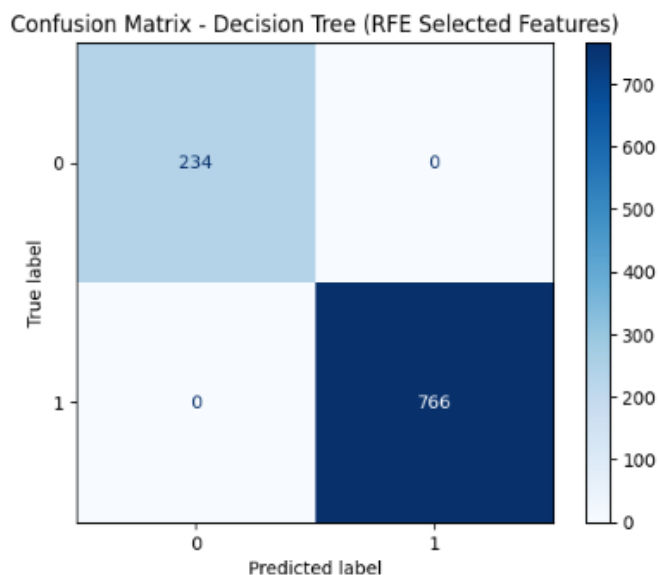
Figure 3.5.5.2

Although the Random Forest model achieved 100% accuracy on the test set, this does not necessarily mean that the model is overfitting. The dataset contains very strong predictors, especially late payment count and credit score, which together account for more than 80% of the model's decisions. These features clearly separate defaulting customers from non-defaulting customers, allowing the model to perform very well on unseen data.

However, achieving 100% accuracy is uncommon in real-world credit-risk modelling. Therefore, the results should be interpreted carefully. The perfect performance may mainly be due to the strong predictive information in the dataset rather than being overfitted by the model (Lou, 2024).

### 3.5.6 Decision Tree

#### Confusion Matrix



**Figure 3.5.6.1**

The Decision Tree achieved perfect classification, correctly predicting all 234 class-0 cases and all 766 class-1 cases. This shows that the selected RFE features made it easy for the model to clearly distinguish between the two classes without making any errors.

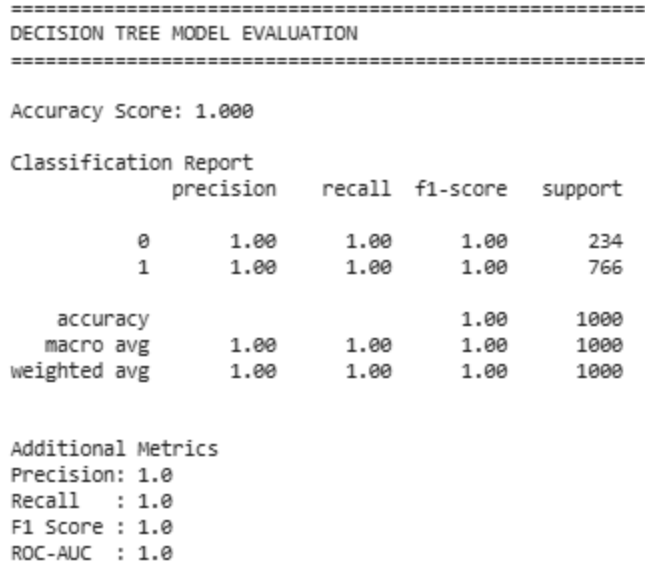


Figure 3.5.6.2

The Decision Tree model achieved perfect results across all evaluation measures, including accuracy, precision, recall, F1-score, and ROC-AUC. Although perfect results are uncommon in real-world credit-risk modelling, they may be explained by the characteristics of this dataset.

The ANOVA F-score results show that late payment count and credit score are very strong predictors of loan default. Together, these features explain more than 80% of the variation in default behavior. They create a clear difference between defaulting and non-defaulting customers, which allowed the Decision Tree to classify all test cases correctly.

Therefore, the perfect performance on the test set may be mainly due to the strong predictive information in the dataset rather than overfitting. However, this result should still be interpreted with caution because perfect performance is unusual in real-world data (Lou, 2024).

3.5.7 Gradient Boosting (XGBoost)

Confusion Matrix

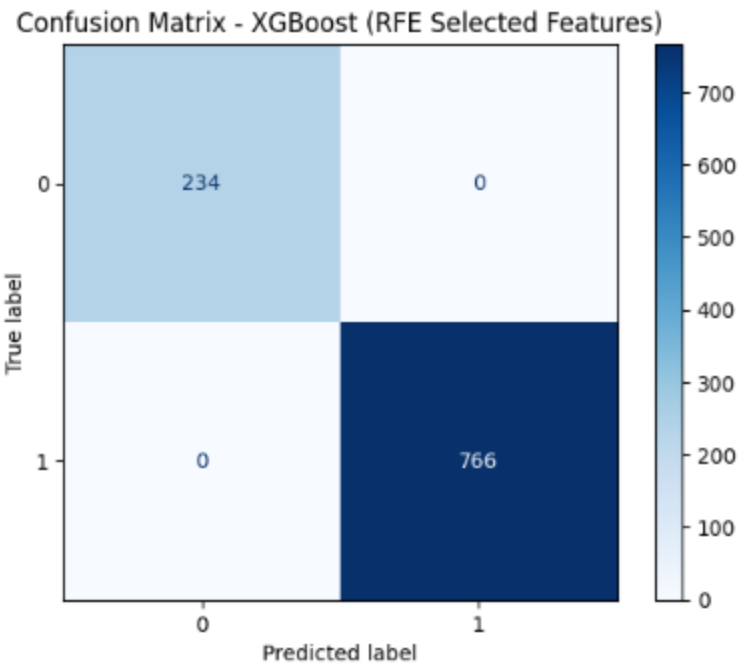


Figure 3.5.7.1

XGBoost achieved perfect classification, correctly predicting all cases in the test set. The model used the RFE-selected features effectively, allowing it to clearly distinguish between the two classes without making any errors.

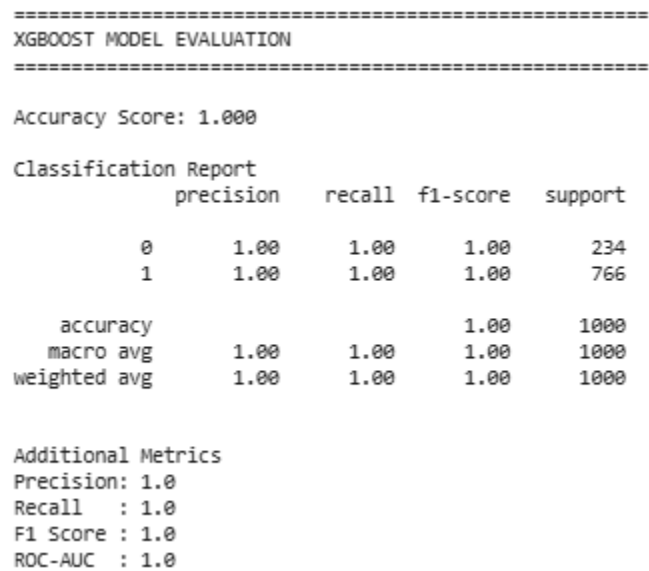


Figure 3.5.7.2

The XGBoost model achieved perfect results across all evaluation measures, including accuracy, precision, recall, F1-score, and ROC-AUC. Although perfect results are

uncommon in real-world credit-risk modelling, they may be explained by the characteristics of this dataset.

The ANOVA F-score results show that late payment count and credit score are very strong predictors of loan default. Together, these features explain more than 80% of the variation in default behavior. They create a clear difference between defaulting and non-defaulting customers, which allowed the XGBoost model to classify all test cases correctly.

Therefore, the perfect performance on the test set may mainly be due to the strong predictive information in the dataset rather than overfitting. However, the results should still be interpreted with caution because perfect performance is unusual in real-world credit-risk modelling (Lou, 2024).

## Conclusion

Four supervised learning models were evaluated to predict loan default: Logistic Regression, Decision Tree, Random Forest, and XGBoost. All models were trained using the same RFE-selected features to ensure a fair comparison. Their performance was evaluated using accuracy, precision, recall, F1-score, and ROC-AUC.

Three models, Decision Tree, Random Forest, and XGBoost, achieved perfect results on the test set, with accuracy, precision, recall, F1-score, and ROC-AUC all equal to 1.000. Logistic Regression also performed very well, with an accuracy of approximately 98.9%, although its precision and recall were slightly lower.

The perfect results achieved by the tree-based models are unusual in real-world credit-risk modelling. However, they may be explained by the characteristics of this dataset. The ANOVA F-score results show that late payment count and credit score are very strong predictors of loan default. Together, these features explain more than 80% of the variation in default behaviour. They create a clear difference between defaulting and non-defaulting customers, allowing the tree-based models to classify all test cases correctly.

Overall, the results show that the dataset contains very strong predictors of loan default. However, the perfect performance should be interpreted carefully, as such results may not be expected when the models are applied to more complex real-world credit-risk data.

XGBoost was selected as the final model for this study. Although the Decision Tree and Random Forest models also achieved perfect performance on the test set, XGBoost was selected because of its boosting approach, which can capture complex and nonlinear relationships between variables. XGBoost is also widely used for structured data and credit-risk modelling. Therefore, based on its modelling capabilities and

suitability for the research problem, XGBoost was selected as the final model for further analysis.

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